TTDS 2024 Past Paper Deep Solutions

Original-question guide + worked solutions

Generated for TTDS revision

2026-05-20

Contents

使用方式

这份解析对应 `2024462_INFR11229.pdf`。重点是 TTDS 经典 calculation questions：`IR evaluation`、`tf-idf`、`language modelling`、`Rocchio`、`mutual information`。所有关键术语保留英文。

Question 1: Preprocessing, Crawling, Permuterm, Compression

Original question screenshot

Please provide answers to **ALL** of the following questions. It is more important to have a clear answer than a longer answer. When calculations are needed, please **show steps** and give answer to **three** decimal digits.

1. (a) For each of the following preprocessing steps, explain the step, and list an advantage and a drawback with an example of each situation: [*3 marks*]

   i. Stemming

   ii. Removing non-alphanumeric characters

   (b) How do web search engines crawl the web? [*2 marks*]

   (c) What is a Permuterm Index? What is the main drawback of it? What should be the lookup query for "*tion" and for "blo*k"? [*2 marks*]

   (d) Politeness and Freshness are the main two considerations that web crawling should consider.

   i. How to balance between these two considerations? And why is that important? [*2 marks*]

   (e) If the following bit sequence is using v-encoding for a delta-encoded index, what is the corresponding delta-encoding for it? What is the original posting list (assuming the first posting stays the same)? [*1 mark*]

   10010010 00100011 00100100 10000001 00001001 10000101

原题要点

考 stemming、removing non-alphanumeric characters、web crawling、permuterm index、politeness vs freshness、v-byte decoding + delta decoding。

(a) Preprocessing

Stemming
Definition: reduce inflected/derived words to a common stem, e.g. `connect`, `connected`, `connection` -> `connect`.

Advantage: improves recall by matching morphological variants.
Drawback: can reduce precision through overstemming, e.g. `universe` and `university` may be conflated by aggressive stemmers.

Removing non-alphanumeric characters
Definition: remove punctuation, symbols, emojis, URLs fragments, etc.

Advantage: reduces noise and vocabulary size, useful for clean news-style corpora.
Drawback: can destroy useful meaning, e.g. `C++`, `COVID-19`, hashtags, email addresses, emoticons, or sentiment punctuation.

(b) How web search engines crawl

Web crawlers start from seed URLs, fetch pages, parse links, normalise URLs, add unseen URLs to a URL frontier, obey `robots.txt` and politeness constraints, revisit pages for freshness, and store downloaded pages for indexing.

(c) Permuterm Index

`Permuterm index` supports wildcard queries by storing all rotations of each term plus an end marker `$`.

Example for `hello$`: `hello$`, `ello$h`, `llo$he`, `lo$hel`, `o$hell`, `$hello`.

Main drawback: dictionary size blows up because each term of length m creates $m + 1$ rotations.

Wildcard lookup:

- `*tion` -> rotate to `tion$*`, look up prefix `tion$`
- `blo*k` -> add `$`: `blo*k$`; rotate so wildcard at end -> `k$blo*`, look up prefix `k$blo`

(d) Politeness vs Freshness

`Politeness` means not overloading servers; `freshness` means revisiting pages often enough to keep the index up to date. Balance them by per-host crawl delays/back queues, priority scores, and adaptive revisit frequency: fast-changing important pages get revisited more often, but same-host requests are spaced out.

(e) v-byte and delta decoding

Sequence:

`10010010 00100011 00100100 10000001 00001001 10000101`

Using lecture convention: high bit 1 terminates a number, high bit `0` continues, remaining 7 bits are data.

Groups:

- `10010010` -> data `0010010` = 18
- `00100011 00100100 10000001` -> data `0100011 0100100 0000001` = 578049
- `00001001 10000101` -> data `0001001 0000101` = 1157

Delta/gap list:

$$[18,\ 578049,\ 1157]$$

Original postings list, assuming first posting unchanged:

$$[18,\ 18 + 578049,\ 578067 + 1157] = [18,\ 578067,\ 579224]$$

常见坑

`v-byte` 先解出 gaps，不是直接得到 postings list。恢复 postings list 要做 cumulative sum。

Question 2: IR Evaluation

Original question screenshot

2. A collection of 10,000 documents has been searched by systems S1 to S5 using query Q. Each system has returned the entire collection of 10,000 documents as a ranked list. The following table shows the top 20 documents returned by each system. Documents A, B, C, D, E, F, G and H are relevant. The letter N is used to denote the positions of non-relevant documents in the search results.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
S1	D	A	N	N	F	N	H	E	N	N	C	N	N	G	B	N	N	N	N	N
S2	N	E	N	H	A	C	N	N	N	N	D	N	F	N	N	N	N	N	N	N
S3	A	N	C	N	B	D	N	N	N	N	E	N	F	N	N	G	N	N	N	H
S4	E	C	N	A	N	D	N	B	N	N	N	N	H	N	N	N	N	F	G	N
S5	C	N	D	A	B	N	N	N	N	N	N	N	F	N	N	G	N	N	H	E

(a) Calculate each of the following metrics for each of the systems, then rank the systems by each of the metrics. Report your result in the format $S1 > S2 > S3 = S4 > S5$ where $S1 > S2$ means that system 1 performs better than system 2, and $S3 = S4$ means they perform the same. [*3 marks*]

i. Precision at R=50%
ii. R@6
iii. AP

(b) For each system, calculate the F1 score on the top 10 documents [*2 marks*]

(c) What is the largest possible MAP that the system S2 could have? Explain [*1 mark*]

(d) Assuming the graded relevance (gain) of documents are as follows: **A**=**B**=**3**, **C**=**D**=**E**=**2**, and **F**=**G**=**H**=**1**. What is the nDCG@10 for system S2. Show your steps. [*2 marks*]

(e) Assuming the same graded relevance of documents as above. Which system has the highest nDCG@1? Which one has the highest nDCG@2? [*2 marks*]

原题要点

Systems S1-S5 return top 20. Relevant docs are A-H. Need:

- `Precision at R=50%`
- R@6
- `AP`
- `F1@10`
- largest possible MAP/AP for S2

- `nDCG@10` for S2 with graded relevance
- highest `nDCG@1` and `nDCG@2`

Relevant ranks

System	Relevant ranks
S1	1, 2, 5, 7, 8, 11, 14, 15
S2	2, 4, 5, 6, 11, 13
S3	1, 3, 5, 6, 11, 13, 16, 20
S4	1, 2, 4, 6, 8, 13, 18, 19
S5	1, 3, 4, 5, 13, 16, 19, 20

(a) Metrics and rankings

`Precision at R=50%` means precision when recall first reaches 50%. There are 8 relevant docs, so recall 50% means retrieving 4 relevant docs.

System	P at R=50%	R@6	AP
S1	0.571	0.375	0.672
S2	0.667	0.500	0.398
S3	0.667	0.500	0.586
S4	0.667	0.500	0.664
S5	0.800	0.500	0.593

Rankings:

- P at R=50%: S5 > S2 = S3 = S4 > S1
- R@6: S2 = S3 = S4 = S5 > S1
- AP: S1 > S4 > S5 > S3 > S2

AP formula:

$$AP = \frac{1}{R}\sum_{k=1}^{n} P@k \cdot rel(k)$$

(b) F1@10

Top 10 relevant counts:

- S1: 5 relevant -> $P = 5/10 = 0.5$, $R = 5/8 = 0.625$, $F1 = 0.556$
- S2: 4 relevant -> $P = 0.4$, $R = 0.5$, $F1 = 0.444$
- S3: 4 relevant -> $F1 = 0.444$
- S4: 5 relevant -> $F1 = 0.556$
- S5: 4 relevant -> $F1 = 0.444$

System	F1@10
S1	0.556
S2	0.444
S3	0.444
S4	0.556
S5	0.444

(c) Largest possible AP/MAP for S2

The table says top 20 only, but each system returned the whole collection. For S2, relevant docs shown are E, H, A, C, D, F: 6 out of A-H. Missing relevant docs B and G could appear immediately at ranks 21 and 22 in the best case.

Then S2 relevant ranks are:

$$2, 4, 5, 6, 11, 13, 21, 22$$

$$AP_{max} = \frac{1}{8}\left(\frac{1}{2} + \frac{2}{4} + \frac{3}{5} + \frac{4}{6} + \frac{5}{11} + \frac{6}{13} + \frac{7}{21} + \frac{8}{22}\right) = 0.485$$

For a single query, MAP equals AP. So largest possible MAP for S2 on this query is 0.485.

(d) nDCG@10 for S2

Graded relevance: A=B=3, C=D=E=2, F=G=H=1.

S2 top 10:

`N, E, N, H, A, C, N, N, N, N`

Gains:

$$[0, 2, 0, 1, 3, 2, 0, 0, 0, 0]$$

Using:

$$DCG@10 = \sum_{i=1}^{10} \frac{gain_i}{\log_2(i+1)}$$

$$DCG@10 = 3.566$$

Ideal gains:

$$[3, 3, 2, 2, 2, 1, 1, 1, 0, 0]$$

$$IDCG@10 = 8.533$$

$$nDCG@10 = 3.566/8.533 = 0.418$$

(e) Highest nDCG@1 and nDCG@2

For `nDCG@1`, only first document matters. S3 has A at rank 1, gain 3, so S3 has nDCG@1 = 1. S1/S4/S5 have gain 2 at rank 1, so 0.667. S2 has N, so 0.

Highest nDCG@1: **S3**.

For `nDCG@2`, S1 ranks D,A gains [2,3], S4 ranks E,C gains [2,2], S5 ranks C,N gains [2,0], S3 ranks A,N gains [3,0], S2 ranks N,E gains [0,2]. Highest is **S1**.

Question 3: tf-idf, Jaccard, LM, Rocchio

Original question screenshot

3. The query "brown fox" is run against the following collection of 8 documents:

d1: brown fox brown fox
d2: run watch fox jump
d3: fat brown fat fox
d4: go fox brown go
d5: jump brown brown fox
d6: brown brown brown jump
d7: watch jump run fox
d8: nice quite lovely fox

All statistics should be computed over this collection. Use base 10 for logarithms and do not add epsilon values.

(a) Which word (or words) have the highest idf value in this collection? Report the word (or words) and their idf value to three decimal digits. [*2 marks*]

(b) What is the Jaccard coefficient between d3 and d5? [*1 mark*]

(c) In the Language Modelling approach, what is the probability P(fat|d3). Assume Jelinek-Mercer smoothing with $\lambda = 0.5$. Report a number to three decimal digits. Show your work. [*2 marks*]

(d) Rank the documents according to the tf.idf weighted sum score for the query. Do not squash the tf component. Do not normalise for document length. Compute and report idf from the 8 documents (remember as always to show steps). Be sure to indicate ties. For example, if you think d1 would receive the highest score, followed by d2, d3 and d4 with the same score, followed by d5, then followed by d6-d8 all with the same score, then you would answer: [*3 marks*]

$$\text{d1} > \text{d2} = \text{d3} = \text{d4} > \text{d5} > \text{d6} = \text{d7} = \text{d8}$$

(e) The user indicates that d7 is relevant to the query, and d1 is non-relevant. Use the Rocchio algorithm to construct a new query. Set $\alpha = \beta = \gamma = 1$ and use tf.idf weights as in part (d) for all vectors. Do not normalise vectors to unit length. Report the resulting query vector. Show three decimal digits. Discuss the resulting query vector and compare it with the initial one. [*2 marks*]

Page 3 of 5

原题要点

Query: `brown fox`。8 documents. Use base 10 logs, raw tf, no document length normalisation.

Documents:

Doc	Text
d1	brown fox brown fox

Doc	Text
d2	run watch fox jump
d3	fat brown fat fox
d4	go fox brown go
d5	jump brown brown fox
d6	brown brown brown jump
d7	watch jump run fox
d8	nice quite lovely fox

(a) Highest idf

$$idf(t) = \log_{10} \frac{N}{df(t)}, \quad N = 8$$

Terms with $df = 1$ have highest idf:

`fat`, `go`, `nice`, `quite`, `lovely`

$$idf = \log_{10}(8/1) = 0.903$$

(b) Jaccard(d3, d5)

Use sets of terms:

- d3 = {fat, brown, fox}
- d5 = {jump, brown, fox}

Intersection = {brown, fox}, size 2. Union = {fat, jump, brown, fox}, size 4.

$$Jaccard = \frac{2}{4} = 0.500$$

(c) Language modelling: $P(fat|d3)$ with Jelinek-Mercer

$$P(t|d) = \lambda P_{MLE}(t|d) + (1 - \lambda)P(t|C)$$

$\lambda = 0.5.$

In d3, `fat` appears 2 times out of 4:

$$P_{MLE}(fat|d3) = 2/4 = 0.5$$

Collection has 32 tokens total; `fat` appears 2 times:

$$P(fat|C) = 2/32 = 0.0625$$

$$P(fat|d3) = 0.5(0.5) + 0.5(0.0625) = 0.281$$

(d) tf-idf ranking for query `brown fox`

Relevant idf values:

$$idf(brown) = \log_{10}(8/5) = 0.204$$

$$idf(fox) = \log_{10}(8/7) = 0.058$$

Score:

$$score(d,q) = tf(brown,d)idf(brown) + tf(fox,d)idf(fox)$$

Doc	Score
d6	0.612
d1	0.524
d5	0.466
d3	0.262
d4	0.262
d2	0.058
d7	0.058
d8	0.058

Ranking:

$$d6 > d1 > d5 > d3 = d4 > d2 = d7 = d8$$

(e) Rocchio feedback

User marks d7 relevant and d1 non-relevant. Set $\alpha = \beta = \gamma = 1$.

$$\vec{q}' = \vec{q}_0 + \vec{d}_{rel} - \vec{d}_{nonrel}$$

Initial query vector:

$$\vec{q}_0 = \{brown : 0.204,\ fox : 0.058\}$$

d7 = `watch jump run fox`:

$$\vec{d}_7 = \{watch : 0.602,\ jump : 0.301,\ run : 0.602,\ fox : 0.058\}$$

d1 = `brown fox brown fox`:

$$\vec{d}_1 = \{brown : 0.408,\ fox : 0.116\}$$

Thus:

Term	New weight
brown	-0.204
fox	0.000
jump	0.301
run	0.602
watch	0.602
all other terms	0.000

Interpretation: feedback moves the query away from d1's repeated `brown fox` and toward d7's terms `watch`, `run`, `jump`. Negative weights are sometimes clipped in practical systems.

Question 4: Expected Mutual Information

Original question screenshot

4. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Cooking* or *Business*. You have identified a number of words that are of interest to you, such as *stocks* and *tree*, and counted how many documents in each category they appear in:

	Animals	Cooking	Tech	Arts	Business	Entertainment
Total no. of docs	2303	1361	803	1403	650	801
stocks	12	6	120	12	278	56
llama	452	24	320	41	3	3
biden	34	31	75	7	107	43
apple	56	156	310	23	2	41
alpaca	313	41	149	26	0	6
pear	4	167	0	3	5	107
tree	102	89	51	102	16	107

$$I(U;C) = \sum_{e_t\in\{1,0\}} \sum_{e_c\in\{1,0\}} P(U=e_t, C=e_c)\log_2 \frac{P(U=e_t, C=e_c)}{P(U=e_t)P(C=e_c)}$$

$$= \frac{N_{11}}{N}\log_2\frac{NN_{11}}{N_{1.}N_{.1}} + \frac{N_{01}}{N}\log_2\frac{NN_{01}}{N_{0.}N_{.1}} + \frac{N_{10}}{N}\log_2\frac{NN_{10}}{N_{1.}N_{.0}} + \frac{N_{00}}{N}\log_2\frac{NN_{00}}{N_{0.}N_{.0}}$$

(a) Calculate the expected mutual information $I(U;C)$ for the word *apple* and the category *Tech* (show steps).

(b) Calculate the expected mutual information $I(U;C)$ for *llama* and *Tech*.

(c) Which of the two words, *apple* and *llama*, occurs more often in documents of the category *Tech*? Which one has the higher expected mutual information? Briefly comment on your observation: if it is the same word for both questions, why do you think that is? If one word occurs more often but the other one has higher MI, why?

[*8 marks*]

5. (a) Describe the process of manual content analysis.

(b) Give an example of a situation where it makes sense to automate this process using a text classifier, and explain why.

(c) Give an example of a situation where it does not make sense to automate this process, and explain why. [*3 marks*]

6. Provide a short answer to each of the following questions:

(a) What is the difference between the unigram model and the mixture of unigrams model?

QUESTION CONTINUES ON NEXT PAGE

Page 4 of 5

原题要点

Compute MI for `apple` with `Tech`, and `llama` with `Tech`, then compare frequency vs MI.

Totals:

$$N = 2303 + 1361 + 803 + 1403 + 650 + 801 = 7321$$

(a) apple and Tech

For `apple`:

- $N_{11} = 310$
- $N_{1.} = 56 + 156 + 310 + 23 + 2 + 41 = 588$
- $N_{.1} = 803$
- $N_{10} = 278$
- $N_{01} = 493$
- $N_{00} = 6240$

Contributions:

Cell	Contribution
N_{11}	0.095910
N_{01}	-0.039261
N_{10}	-0.034674
N_{00}	0.049357

$$I(apple; Tech) = 0.071$$

(b) llama and Tech

For `llama`:

- $N_{11} = 320$
- $N_{1.} = 452 + 24 + 320 + 41 + 3 + 3 = 843$
- $N_{.1} = 803$
- $N_{10} = 523$
- $N_{01} = 483$
- $N_{00} = 5995$

Contributions:

Cell	Contribution
N_{11}	0.078289
N_{01}	-0.036740
N_{10}	-0.037227
N_{00}	0.045712

$$I(llama; Tech) = 0.050$$

(c) Comparison

`llama` occurs slightly more often in Tech documents: 320 vs apple 310. But `apple` has higher MI: 0.071 vs 0.050.

Reason: MI measures association, not just raw within-class frequency. `llama` appears often in Animals too, so it is less specifically associated with Tech. `apple` is more distinctive for Tech relative to its overall distribution.

Question 5: Manual Content Analysis and Automation

Original question screenshot

4. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Cooking* or *Business*. You have identified a number of words that are of interest to you, such as *stocks* and *tree*, and counted how many documents in each category they appear in:

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$$= \frac{N_{11}}{N} \log_2 \frac{NN_{11}}{N_{1.}N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{NN_{01}}{N_{0.}N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{NN_{10}}{N_{1.}N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{NN_{00}}{N_{0.}N_{.0}}$$

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[*8 marks*]

5. (a) Describe the process of manual content analysis.

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(c) Give an example of a situation where it does not make sense to automate this process, and explain why. [*3 marks*]

6. Provide a short answer to each of the following questions:

(a) What is the difference between the unigram model and the mixture of unigrams model?

QUESTION CONTINUES ON NEXT PAGE

(a) Manual content analysis

Manual content analysis means researchers define a coding scheme/categories, sample documents, train human coders, annotate documents, check inter-annotator agreement, refine coding guidelines, then analyse counts/patterns.

(b) When automation makes sense

Example: classifying millions of tweets as customer complaints vs praise. It makes sense because scale is high, labels are relatively consistent, and small errors may be acceptable for aggregate analysis.

(c) When automation does not make sense

Example: high-stakes legal judgement or subtle qualitative interpretation of political rhetoric. It may not make sense because context is complex, labels require expert judgement, and errors have serious consequences.

Question 6: Topic Modelling and Classification Short Answers

Original question screenshot

4. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Cooking* or *Business*. You have identified a number of words that are of interest to you, such as *stocks* and *tree*, and counted how many documents in each category they appear in:

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(a) Calculate the expected mutual information $I(U;C)$ for the word *apple* and the category *Tech* (show steps).

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(c) Give an example of a situation where it does not make sense to automate this process, and explain why. [*3 marks*]

6. Provide a short answer to each of the following questions:

(a) What is the difference between the unigram model and the mixture of unigrams model?

QUESTION CONTINUES ON NEXT PAGE

(b) Why is the generative process for LDA unrealistic? Use an example sentence to illustrate your point.

(c) i. Explain the term "out-of-vocabulary words" in the context of text classification.

ii. Name one approach we can use to deal with out-of-vocabulary words in traditional text classification (e.g. when training an SVM model from scratch).

iii. Explain briefly why this is less of an issue with transfer learning approaches.

(d) What is zero-shot classification and why is it useful? *[6 marks]*

7. Your colleague is building a search engine. She has put a great deal of effort into hand-crafting features, but she is not quite happy with the results yet. Name and describe three ways that query logs and click-through data might help her improve her search engine. *[3 marks]*

(a) Unigram model vs mixture of unigrams

`Unigram model` assumes all words in a document are generated from one word distribution. `Mixture of unigrams` assumes each document first chooses one latent topic/component, then all words are generated from that topic's word distribution. It captures document-level clustering but not multiple topics per document.

(b) Why LDA generative process is unrealistic

LDA assumes each word's topic is sampled independently from the document-topic distribution, ignoring syntax and word order. Example: in "the bank raised interest rates", words are not independent; grammar and collocations matter. LDA is useful as a bag-of-words topic model, not a realistic model of human text production.

(c) OOV and transfer learning

i. OOV words
`Out-of-vocabulary words` are words appearing at test time that were not in the training vocabulary.

ii. Traditional solution
Use feature pruning with an `<UNK>` token, character n-grams, subword features, or retrain vocabulary with more data.

iii. Why less issue with transfer learning
Modern pretrained models often use subword tokenisation such as BPE/WordPiece, so unseen words can be decomposed into known subword units. They also bring pretrained semantic representations.

(d) Zero-shot classification

`Zero-shot classification` means classifying examples into labels without task-specific labelled training examples for those labels. It is useful when labelled data is scarce, classes change often, or we want rapid deployment using natural-language label descriptions.

Question 7: Query Logs and Click-through Data

Original question screenshot

QUESTION CONTINUED FROM PREVIOUS PAGE

(b) Why is the generative process for LDA unrealistic? Use an example sentence to illustrate your point.

(c) i. Explain the term "out-of-vocabulary words" in the context of text classification.

ii. Name one approach we can use to deal with out-of-vocabulary words in traditional text classification (e.g. when training an SVM model from scratch).

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(d) What is zero-shot classification and why is it useful? [*6 marks*]

7. Your colleague is building a search engine. She has put a great deal of effort into hand-crafting features, but she is not quite happy with the results yet. Name and describe three ways that query logs and click-through data might help her improve her search engine. [*3 marks*]

Page 5 of 5

原题要点

Colleague has hand-crafted ranking features but wants improvement. Name three ways query logs/click-through can help.

Answer

1. **Query expansion / suggestion**
   Logs reveal common reformulations and related queries. If users search "car insurance" then "cheap auto insurance", the system can learn query similarity.
2. **Learning-to-rank features**
   Clicks, dwell time, skips, and reformulations can become implicit relevance signals. These train or tune ranking models beyond hand-crafted features.
3. **Detect popular intents and navigational targets**
   For frequent queries, click-through identifies preferred URLs or dominant user intent. This improves ranking for head queries and navigational queries.

Caveats

Click data is biased by rank position and presentation. A top result gets more clicks even if not best, so logs need debiasing or careful interpretation.

Coverage Checklist

Question	Covered TTDS topics
Q1	Preprocessing, stemming, URL/symbol handling, crawling, politeness/freshness, permuterm index, v-byte/delta encoding
Q2	Precision at recall, recall@k, AP/MAP, F1, nDCG, graded relevance
Q3	tf-idf, idf, Jaccard, language modelling, Jelinek-Mercer smoothing, Rocchio relevance feedback
Q4	Expected mutual information, feature selection, contingency tables
Q5	Manual content analysis, automated classification, when ML is appropriate
Q6	Unigram model, mixture of unigrams, LDA assumptions, OOV, transfer learning, zero-shot classification
Q7	Query logs, click-through data, implicit feedback, learning to rank